

2.4 Resource Security

1. The Concept of Resource Security

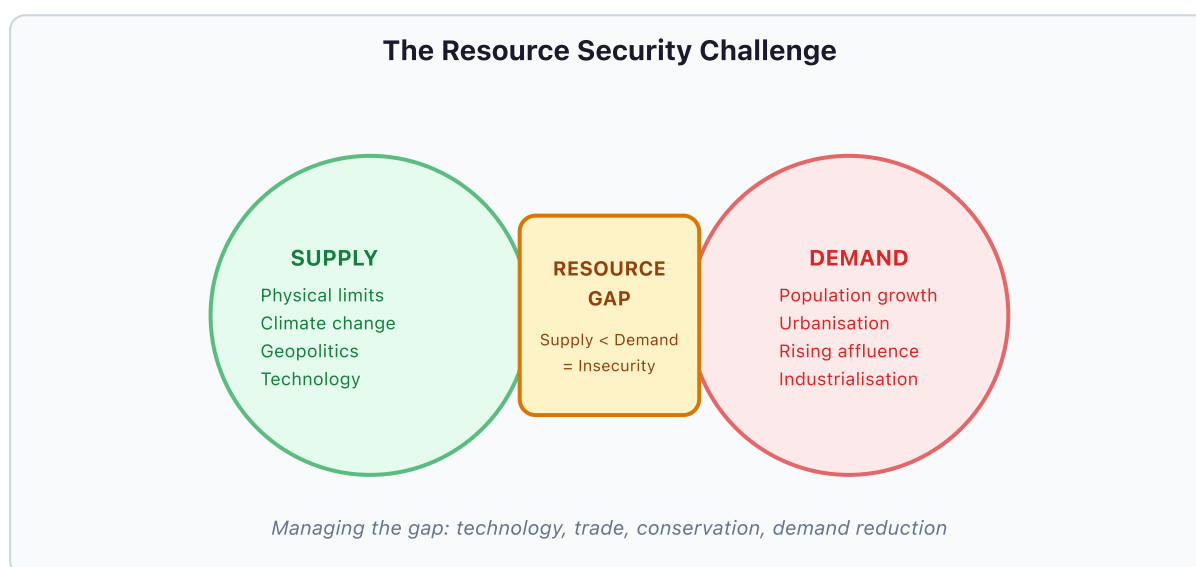
Resource: Anything useful to people that can be extracted from the environment to meet human needs and wants. Resources can be *renewable* (replenished within a human lifetime: solar, wind, timber, water) or *non-renewable* (finite, formed over geological time: fossil fuels, minerals, ores). Some resources are *flow resources* (continuously available regardless of use: solar radiation, tidal energy).

Resource security: The ability of a country or population to access sufficient, affordable, and reliable resources to meet current and future needs. Insecurity occurs when supply cannot meet demand, leading to shortages, price volatility, conflict, and threats to well-being. The concept encompasses energy security, water security, food security, and mineral security.

Factors Affecting Resource Supply and Demand

Supply Factors	Demand Factors
Physical availability (geology, climate, hydrology, soil fertility)	Population growth (8 billion people, rising to ~10 billion by 2060)
Technology for extraction and processing (fracking, desalination, GMOs)	Economic development and industrialisation (rising GDP = rising resource use)
Cost of extraction and transport (determines economic viability)	Rising living standards and changing consumption patterns (meat-rich diets, car ownership)
Political stability and governance (conflict disrupts supply)	Urbanisation (concentrated, intensified demand)

Trade relationships, infrastructure, and supply chains	Technological change (new uses: lithium for batteries, rare earths for electronics)
Environmental regulations and sustainability constraints	Cultural and dietary shifts (e.g. rising meat consumption in China and India)
Climate change impacts on availability (droughts, floods, glacier melt)	Waste and inefficiency in resource use (1/3 of food produced is wasted)



2. Energy Security

Energy security: The uninterrupted availability of energy sources at an affordable price (IEA definition). Encompasses short-term security (ability to react to sudden supply disruptions) and long-term security (timely investment to supply energy in line with economic development and environmental sustainability needs). No country is fully energy secure — all face trade-offs between cost, reliability, and environmental impact.

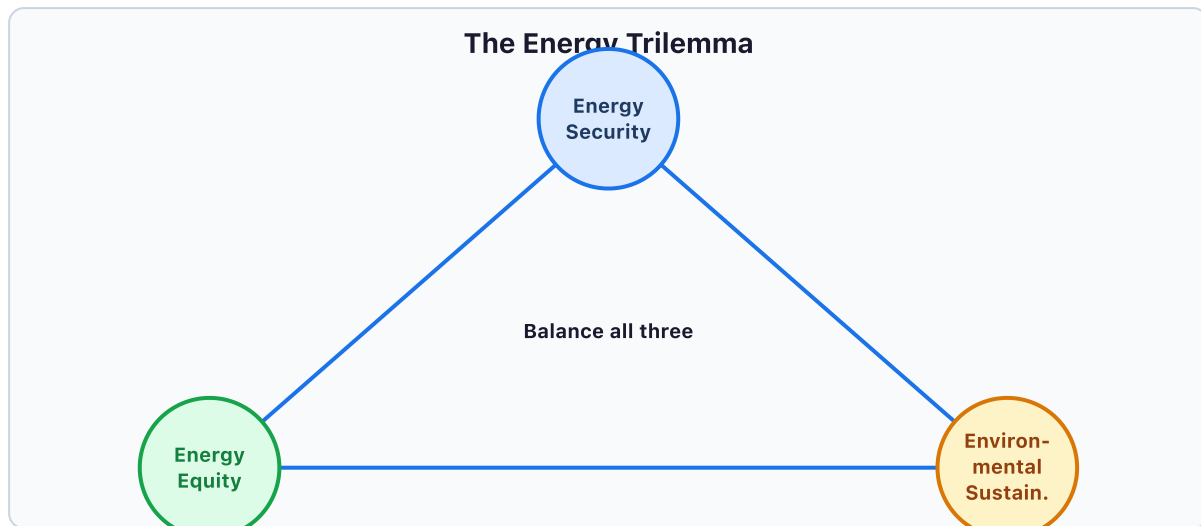
Global Energy Mix

The global energy mix remains dominated by fossil fuels (~80% of primary energy), though the transition to renewables is accelerating:

Source	% Global Primary Energy (2023)	Trend
Oil	~31%	Slowly declining share but absolute use near peak. Transport is ~56% of oil demand. Petrochemicals growing.
Coal	~26%	Declining in Europe/N. America, still growing in India, Indonesia. Provides 36% of global electricity.
Natural gas	~23%	Growing as "transition fuel" (50% less CO ₂ than coal per unit of energy). LNG trade expanding rapidly.
Hydroelectric	~7%	Mature technology. Largest source of renewable electricity. Limited growth potential in HICs; expanding in LICs (e.g. GERD in Ethiopia).
Nuclear	~4%	Stable globally. Declining in Germany (phased out 2023), growing in China (23 reactors under construction).
Wind	~4%	Fastest-growing source after solar. Global capacity doubled between 2018 and 2023. Offshore wind growing rapidly (UK is world leader).
Solar	~2%	Exponential growth: costs fell 90% since 2010. Solar is now the cheapest source of new electricity in most of the world.
Bioenergy/other	~3%	Includes traditional biomass (cooking fuel in LICs), modern biofuels, geothermal, tidal.

The Energy Trilemma

Energy policy must balance three competing priorities, known as the **energy trilemma** (World Energy Council):



1. **Energy security** — reliable, uninterrupted supply. Requires diverse sources, strategic reserves, robust infrastructure, and domestic production or secure imports.
2. **Energy equity (affordability)** — energy must be accessible and affordable for all. Fuel poverty affects 13% of UK households (3.2 million homes). 675 million people globally lack access to electricity.
3. **Environmental sustainability** — minimising carbon emissions, air pollution, and ecosystem damage. Paris Agreement targets require cutting emissions by 43% by 2030 and reaching net zero by 2050.

Factors Affecting Energy Security

- **Physical factors** — geology determines fossil fuel deposits; climate determines solar/wind potential; hydrology determines hydropower capacity.
- **Geopolitics** — OPEC+ controls ~40% of global oil production and can manipulate prices through production quotas. Russia supplied ~40% of Europe's gas before 2022. Concentration of supply creates vulnerability. Pipeline politics: Nord Stream 1 and 2 (Russia-Germany), TAPI pipeline (Turkmenistan-Afghanistan-Pakistan-India).
- **Economic factors** — price volatility (oil prices ranged from -\$37/barrel in April 2020 to \$130/barrel in March 2022), investment cycles, affordability of transition to renewables.
- **Environmental constraints** — climate change mitigation requires rapid decarbonisation, carbon taxes, emission trading schemes, and phase-out of coal and eventually oil/gas.
- **Technology** — fracking unlocked vast shale gas/oil reserves (US became world's largest oil producer in 2018), battery storage enables renewable intermittency management, nuclear fusion research (potentially limitless clean energy), smart grids, and hydrogen fuel cells.

Case Study: The UK's Energy Security Challenge

The UK faces a complex energy transition. **Declining domestic production:** North Sea oil and gas production peaked around 1999; the UK became a net energy importer in 2004. Now imports ~50% of gas (mainly from Norway via pipeline and global LNG markets) and ~80% of oil (refined products). **2022 energy crisis:** Russia's invasion of Ukraine triggered European gas supply disruption; UK wholesale gas prices surged 400%, causing a cost-of-living crisis. Government responded with an energy price cap, £37 billion in household support, and a windfall tax on oil company profits. **Renewable transition:** renewables now generate ~42% of UK electricity (2023), with wind as the single largest source (~27%). Offshore wind capacity: 14 GW (world's second largest after China), targeting 50 GW by 2030. **Controversies:** debate over new North Sea oil licences (Rosebank field approved 2023 despite climate concerns), nuclear new-build (Hinkley Point C delayed to 2029 and over budget at £33+ billion; Sizewell C approved), fracking (banned in England since 2019 moratorium after induced seismicity near Blackpool), onshore wind planning restrictions (effectively banned in England 2015–2023, partially lifted). **Energy poverty:** 3.2 million households in fuel poverty. The energy trilemma is stark: how to keep energy affordable while meeting net zero by 2050 and maintaining security of supply.

Case Study: China's Energy Strategy

China is the world's largest energy consumer and CO₂ emitter (~31% of global emissions). Coal provides ~55% of China's primary energy and ~60% of electricity. However, China is also the world's largest investor in renewable energy: ~\$758 billion invested in clean energy in 2023, more than the rest of the world combined. China dominates solar panel manufacturing (80% of global production), produces 60% of the world's wind turbines, and has 23 nuclear reactors under construction. China added more solar capacity in 2023 (217 GW) than the entire cumulative solar capacity of any other country. China's energy strategy reflects its trilemma: rapid industrialisation demands energy security and affordability (hence continued coal use), while severe air pollution (1.2 million premature deaths/yr) and climate commitments (net zero by 2060) drive the renewable transition. China's Belt and Road Initiative exports both coal-fired power plants AND renewable technology to developing countries. The tension between economic growth, energy security, and environmental sustainability is epitomised by China's approach.

3. Water Security

Water security: The capacity of a population to safeguard access to adequate quantities of acceptable-quality water for sustaining human well-being, socioeconomic development, and protection against water-borne pollution and water-related disasters (UN-Water). Water scarcity is defined as $<1,000 \text{ m}^3/\text{person}/\text{year}$; water stress as $<1,700 \text{ m}^3/\text{person}/\text{year}$. By 2025, 1.8 billion people will live in regions with absolute water scarcity.

Global Water Distribution

Water supply is extremely unevenly distributed. Brazil has ~12% of the world's freshwater; the entire Middle East and North Africa has $<1\%$. Two types of water scarcity:

- **Physical water scarcity** — insufficient water resources to meet demand. Affects arid and semi-arid regions: Middle East, North Africa, Central Asia, parts of India, western USA, Australia. Natural aridity compounded by climate change.
- **Economic water scarcity** — water exists but communities lack the infrastructure, investment, or governance to access it. Common in Sub-Saharan Africa, where water sources exist but treatment plants, pipes, wells, and pumps are lacking. 2 billion people lack safely managed drinking water globally (2023).

Factors Affecting Water Security

- **Climate** — rainfall patterns, aridity, seasonal variability, glacier melt (1.9 billion people depend on glacial meltwater, which is declining as glaciers retreat).
- **Population growth** — global water demand has increased 600% over the past century, growing at ~1%/yr. More people = more domestic, agricultural, and industrial demand.
- **Economic development** — industrialisation and rising living standards increase per-capita water use. A US citizen uses ~300 litres/day; a Mozambican uses ~10 litres/day.
- **Agriculture** — irrigation accounts for ~70% of global freshwater withdrawals. The Green Revolution dramatically increased crop yields but at the cost of intensive water use. Flood irrigation wastes 50–60% of water applied.
- **Pollution** — industrial discharge, agricultural runoff (fertilisers, pesticides), untreated sewage, and plastic waste contaminate freshwater sources. 80% of wastewater globally is discharged untreated.
- **Climate change** — altered precipitation patterns, more frequent and intense droughts and floods, glacier retreat, earlier snowmelt, increased evaporation. IPCC projects that for every 1°C of warming,

~7% of the global population faces a 20%+ decrease in renewable water resources.

- **Governance** — water management, pricing policies, infrastructure investment, corruption, and transboundary agreements all affect access and distribution.
- **Virtual water trade** — water embedded in traded goods. Importing food from water-rich countries can alleviate local water stress. The UK imports ~62% of its virtual water (in food and manufactured goods).

Strategies to Improve Water Security

Strategy	Description	Advantages	Disadvantages
Dams and reservoirs	Store water for supply, irrigation, HEP, and flood control	Reliable supply, renewable energy, flood management, recreation	Displacement (Three Gorges Dam displaced 1.3 million people), ecosystem damage (blocking fish migration, altering sediment flow), sedimentation reduces capacity, high cost, evaporation losses in hot climates
Desalination	Removing salt from seawater using reverse osmosis or distillation	Unlimited raw material (seawater), reliable supply regardless of rainfall. Saudi Arabia produces 50% of water from desalination.	Very energy-intensive (3–10 kWh per m ³), expensive (\$0.50–3.00/m ³), brine disposal harms marine ecosystems, high carbon footprint unless powered by renewables
Water transfer schemes	Moving water from surplus to deficit areas via pipelines, canals, or aqueducts	Addresses regional imbalance (e.g. China's South-North Water Transfer: 44.8 billion m ³ /yr capacity, \$79 billion cost)	Extremely expensive, environmental damage to source region, political conflict between regions, evaporation losses
Rainwater harvesting	Collecting and storing rainwater	Low-cost, decentralised, suitable for LICs,	Seasonal and unreliable, limited volume, quality requires treatment, storage space needed

	from roofs or ground surfaces	reduces demand on mains supply	
Water recycling	Treating wastewater for reuse in irrigation, industry, or even drinking	Reduces demand on freshwater, reduces pollution of waterways. Singapore's NEWater meets up to 40% of water demand	Public perception ("toilet to tap"), treatment costs, energy for treatment
Demand management	Metering, leak repair, efficient irrigation (drip systems), pricing, regulation	Cost-effective, reduces waste, changes behaviour. Metering reduces use by ~10–15%.	Requires behaviour change, pricing may disadvantage the poor, leak repair is expensive (UK loses 3 billion litres/day to leaks)
Appropriate technology	Low-cost, locally maintainable solutions for LICs: hand pumps, sand filters, solar stills, fog nets	Affordable, empowers communities, culturally appropriate, low maintenance	Limited scale, may not meet growing urban demand

Case Study: Water Conflict — The Nile Basin

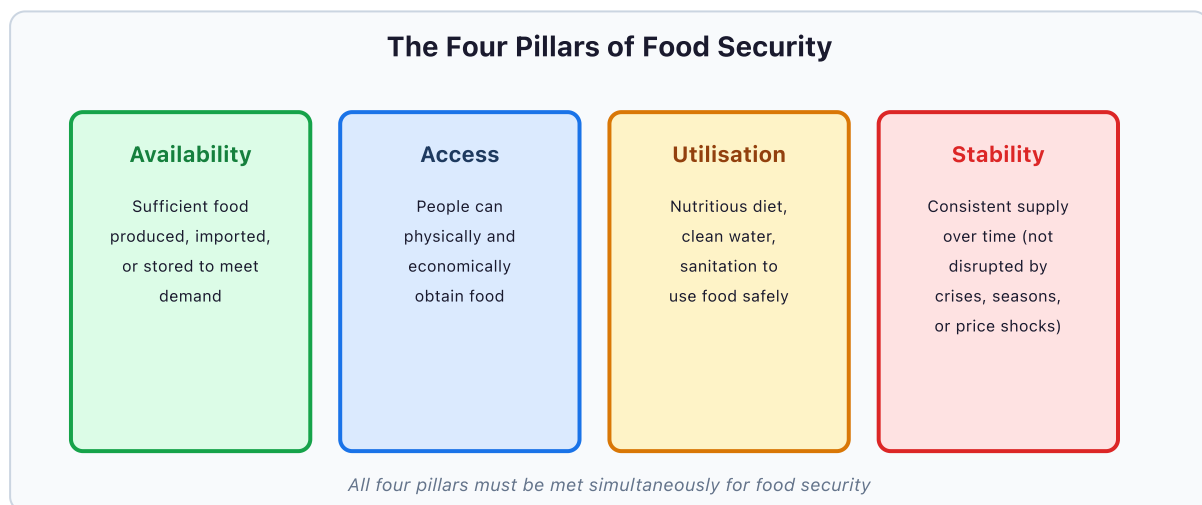
The Nile River (6,650 km, world's longest) flows through 11 countries. **Egypt** depends on the Nile for ~95% of its freshwater and has claimed "historic rights" to 55.5 billion m³/yr since a 1959 treaty with Sudan (which allocated 18.5 billion m³). Ethiopia was not party to this treaty. **Ethiopia's Grand Ethiopian Renaissance Dam (GERD)** on the Blue Nile (which provides 85% of Nile flow) began filling in 2020. The dam's reservoir holds 74 billion m³ — larger than London. Ethiopia argues the dam is essential for development: it will generate 6,000 MW of hydroelectricity (Africa's largest dam), lifting millions from energy poverty (65% of Ethiopians lack electricity). **Egypt's fears:** reduced downstream flow could devastate agriculture (irrigation of 3.5 million hectares), threaten water supply of 105+ million people, reduce Aswan Dam electricity output, and exacerbate Egypt's existing water stress (<500 m³/person/yr, well below the scarcity threshold). **Sudan** has vacillated between supporting and opposing the dam (it would benefit from flood control and cheaper electricity, but risks from dam failure). Negotiations under the African Union have stalled

repeatedly since 2020. The dispute illustrates how transboundary water resources create geopolitical tension, especially when combined with rapid population growth (Ethiopia 126M, Egypt 107M, both growing fast), climate change (projected reduced rainfall in East Africa), and fundamentally different interpretations of water rights (colonial-era treaties vs. sovereign development rights).

4. Food Security

Food security: When all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life (FAO definition, 1996). Food insecurity exists on a spectrum from chronic hunger and undernutrition to micronutrient deficiency ("hidden hunger") and, paradoxically, diet-related obesity.

The Four Pillars of Food Security



Global Food System Challenges

- **Unequal distribution** — global food production is sufficient to feed everyone (~6,000 kcal produced per person per day), but ~735 million people face chronic hunger (2023, FAO) while ~1 billion adults are obese. The problem is access and distribution, not total production.
- **Food waste** — ~1/3 of all food produced globally is lost or wasted (~1.3 billion tonnes/yr, worth ~\$1 trillion). In HICs, waste occurs at retail and consumer level (cosmetic standards, sell-by dates, over-

purchasing). In LICs, waste occurs at production and storage level (lack of refrigeration, poor roads, pest damage, inadequate storage).

- **Climate change** — droughts, floods, heat stress, shifting growing seasons, and pest/disease range expansion threaten crop yields. IPCC projects global crop yields could decline 2–6% per decade due to climate change, while demand increases 14% per decade.
- **Soil degradation** — 33% of global soils are moderately to highly degraded through erosion, salinisation, compaction, acidification, and nutrient depletion. It takes ~500 years to form 2.5 cm of topsoil naturally. UN estimates 24 billion tonnes of fertile soil are lost annually.
- **Water scarcity** — agriculture accounts for 70% of global freshwater use. Aquifer depletion threatens future production (Ogallala Aquifer in US Great Plains has declined 30+ m in parts; North China Plain aquifer falling 1–3 m/yr).
- **Land use competition** — urbanisation, biofuel crops, solar farms, and environmental conservation compete for agricultural land. Globally, 12 million hectares of agricultural land are lost to degradation annually.
- **Supply chain vulnerability** — the COVID-19 pandemic and Russia-Ukraine war (2022) disrupted global food trade, causing price spikes (wheat prices rose 50% in 2022). Ukraine and Russia together supply ~30% of global wheat exports.

Approaches to Food Production

Approach	Description	Advantages	Disadvantages
Industrial/intensive agriculture	Large-scale, mechanised, high-input farming (fertilisers, pesticides, irrigation, monocultures)	High yields per hectare, economies of scale, feeds large populations, low food prices	Environmental damage (soil degradation, water pollution, biodiversity loss, GHG emissions), fossil fuel dependency, reduces rural employment, vulnerable to climate shocks

Green Revolution (1960s–80s)	High-yielding crop varieties (HYVs), irrigation, fertilisers, and pesticides introduced in Asia and Latin America	Doubled cereal yields, prevented mass famine in India and SE Asia, increased food security for billions	Required expensive inputs (disadvantaging small farmers), increased water use, soil degradation, reduced crop diversity, dependency on agrochemicals
GM crops (biotechnology)	Genetically modified organisms engineered for pest resistance, drought tolerance, or nutritional content	Higher yields, reduced pesticide use (Bt cotton), potential for drought/flood-resistant varieties, Golden Rice (Vitamin A enrichment)	Corporate control of seed supply (Monsanto/Bayer), environmental concerns (gene transfer to wild relatives), consumer resistance (especially in EU), expensive technology
Agroecology / organic farming	Working with natural processes: crop rotation, composting, biological pest control, polycultures, no synthetic chemicals	Environmentally sustainable, preserves soil health, enhances biodiversity, lower input costs for small farmers, healthier food	Lower yields per hectare (typically 20–25% less than conventional), more labour-intensive, premium pricing limits access, insufficient alone to feed 10 billion people
Vertical farming / urban agriculture	Growing crops in stacked indoor layers using LED lights, hydroponics, controlled environments	Year-round production, 95% less water than field farming, no pesticides, close to urban consumers (reducing food miles)	Very high energy costs (lighting), limited to leafy greens/herbs (cannot grow staple crops), high capital investment, small scale

Case Study: Sub-Saharan Africa — Food Insecurity

Sub-Saharan Africa has the highest prevalence of food insecurity globally: ~282 million people are undernourished (2023), representing 20% of the population (compared to <2.5% in East Asia).

Causes are complex and interlinked: low agricultural productivity (yields for cereals are 1.5 tonnes/ha vs global average of 4.1 tonnes/ha), poor infrastructure (30–50% of crops lost post-

harvest due to lack of storage and roads), climate variability (droughts in the Horn of Africa, floods in West Africa), conflict (war in Sudan, DRC, Ethiopia destroys food systems), population growth outpacing food production growth (population projected to double to 2.4 billion by 2050), limited access to credit/technology/markets for smallholder farmers (who produce 80% of food), gender inequality (women do 60% of agricultural labour but own only 15% of land), governance failures, and historical legacy of colonialism (export crop dependency over food crops). **Solutions needed:** investment in smallholder agriculture (irrigation, improved seeds, extension services), infrastructure (roads, storage, markets), female empowerment (land rights, education, credit access), climate-resilient farming practices, reduced food waste, improved governance, and addressing conflict. The Alliance for a Green Revolution in Africa (AGRA) promotes improved seeds and fertilisers but has been criticised for promoting a corporate agricultural model over agroecological approaches.

5. Mineral Security

Critical minerals: Minerals essential for modern technology and the green energy transition, whose supply is concentrated in a small number of countries, creating geopolitical vulnerability. Examples: lithium (batteries), cobalt (batteries), rare earth elements (magnets for wind turbines and EVs), copper (wiring), silicon (solar panels), graphite (batteries), nickel, platinum.

The transition from fossil fuels to renewable energy *increases* demand for certain minerals dramatically. An electric car requires 6x more minerals than a conventional car. A single offshore wind turbine contains ~8 tonnes of copper and rare earth magnets. Global demand for lithium is projected to increase 42x by 2040 (IEA). Supply is highly concentrated: DRC produces 70% of global cobalt (often using child labour in artisanal mines); China processes 60% of lithium and 90% of rare earth elements; Chile holds 42% of known lithium reserves. This concentration creates new dependencies that mirror (and may replace) fossil fuel dependencies. The geopolitical implications are profound: China dominates critical mineral processing and could restrict exports as leverage, much as OPEC controls oil supply.

6. The Nexus: Interconnections Between Resources

Energy, water, and food security are deeply interconnected — the **water-energy-food nexus**:

- **Water for energy** — thermal power stations require cooling water; hydropower requires river flow; fracking uses millions of litres per well; biofuel crops need irrigation.
- **Energy for water** — pumping, treating, desalinating, and distributing water all require energy. In many countries, 10–20% of electricity is used for water supply.
- **Water for food** — agriculture uses 70% of global freshwater. It takes ~1,800 litres of water to produce 1 kg of wheat and ~15,400 litres for 1 kg of beef.
- **Energy for food** — modern agriculture depends on fossil fuels for machinery, fertiliser production (Haber-Bosch process uses ~1.2% of global energy), processing, refrigeration, and transport.
- **Food for energy** — biofuels (ethanol from corn/sugarcane, biodiesel from palm oil/rapeseed) compete with food production for land and water.

This means that insecurity in one resource creates cascade effects across the others. Climate change acts as a "threat multiplier," simultaneously affecting water availability, crop yields, and energy demand.

Exam Tip: AQA resource security essays require you to evaluate the relative importance of different factors. For example: "Evaluate the extent to which technology can solve the challenge of water insecurity." Structure: (1) technologies that help (desalination, drip irrigation, water recycling, cloud seeding), (2) limitations of technology (cost, energy requirements, environmental impacts, access inequality between HICs and LICs), (3) non-technological solutions (governance, pricing, demand management, reducing waste, conflict resolution), (4) conclusion weighing both sides. Always use specific named examples with data.

Exam Tip: When discussing resource security, always consider the **scale** of the issue (global, national, local), the **timescale** (short-term crisis vs long-term challenge), and **whose perspective** is being considered (governments, corporations, communities, individuals, future generations). Resource security is inherently political: who controls resources, who benefits, and who suffers are central geographical questions.

Exam Tip: The water-energy-food nexus is a powerful framework for showing synoptic understanding. If asked about water insecurity, discuss how it connects to energy (hydropower depends on water; desalination requires energy) and food (irrigation accounts for 70% of water use; drought reduces crop yields). Showing these interconnections demonstrates the holistic, systems-thinking approach that AQA rewards with top marks.

7. Country Case Studies: Resource Insecurity

Yemen: Food and Water Crisis

Case Study: Yemen — Food and Water Crisis

Yemen exemplifies the catastrophic intersection of conflict, water scarcity, food insecurity, and economic collapse. By 2024, Yemen is the world's worst humanitarian crisis: 21.6 million people (65% of the population) are food insecure; 17 million face acute food insecurity; ~4 million are internally displaced; and an estimated 150,000–375,000 people have died from the conflict (2015–present), with up to 60% of deaths from disease, famine, and lack of healthcare access rather than direct violence.

Water crisis background: Yemen is the Arab world's most water-scarce country with only ~86 m³ of renewable freshwater per person per year (well below the 1,000 m³ scarcity threshold; global average ~5,800 m³). Sana'a, the capital, was predicted to be the first capital city in the world to run out of water (projections ranged from 2015–2025). Underlying causes: rapid population growth (doubled since 1990 to 34 million), extensive unregulated groundwater extraction for qat cultivation (a mildly narcotic shrub chewed recreationally, which accounts for 30–40% of agricultural water use in some areas), climate change reducing rainfall, and deteriorating infrastructure.

Conflict amplification: The Yemen Civil War (2015–present) between the Houthi movement (backed by Iran) and the Saudi-led coalition (backed by the USA, UK, UAE) has systematically destroyed water, food, and healthcare infrastructure. Key supply lines have been blockaded (the Saudi coalition's blockade of Hodeidah port, through which 70% of Yemen's food imports arrived, severely reduced food access). Water treatment and pumping stations have been directly targeted or damaged. The UN has documented use of food as a weapon of war.

Food security dimensions: Yemen imports ~90% of its food (it historically exported incense and coffee but deforestation, land degradation, and rising food imports made it dependent). The combination of import dependency, currency collapse (Yemeni rial lost 90% of its value), loss of export earnings, and conflict-disrupted logistics created hyperinflationary food prices: the cost of a basic food basket increased 400–500% between 2015 and 2023. Agricultural production collapsed (30% below pre-war levels). International food aid (World Food Programme) reaches

millions but is constrained by access negotiations, funding gaps, and logistical barriers.

Geographical significance: Yemen demonstrates how multiple resource insecurities (water, food, energy) interact with governance failure and conflict to create humanitarian catastrophe. It also shows how water scarcity can be both a background structural cause and a consequence of conflict — the water-conflict nexus. The case challenges the idea that resource insecurity is primarily a problem of physical scarcity: Yemen's resource crisis is fundamentally political, not physical.

Qatar: Water Stress and Desalination Dependence

Case Study: Qatar — Water Insecurity in a Wealthy State

Qatar is one of the world's most water-stressed countries, with essentially zero renewable freshwater resources: less than 74 mm/yr rainfall, no permanent rivers, and negligible aquifer recharge. Yet Qatar has among the highest per-capita water consumption in the world (400–500 litres/day compared to the UK average of 140 litres/day) due to its hot, arid climate (water-intensive air conditioning, green spaces, and swimming pools), large expatriate population, and heavily subsidised (near-zero) water prices.

Desalination dependence: Qatar produces >99% of its freshwater through seawater desalination, primarily using Multi-Stage Flash (MSF) and Reverse Osmosis (RO) technologies powered by natural gas. Total desalination capacity: ~1.7 million m³/day. The desalination plants at Ras Abu Fontas are among the largest in the world.

Vulnerabilities:

- *Energy dependency:* desalination is extremely energy-intensive (~3–10 kWh per m³). Qatar's entire water supply depends on continued natural gas availability and functioning power infrastructure. A prolonged power failure or disruption to gas supplies would eliminate water production within days.
- *Strategic vulnerability:* the 2017–2021 Qatar diplomatic crisis (Saudi Arabia, UAE, Bahrain, and Egypt severed diplomatic ties and blockaded Qatar) included a threat to cut water pipelines. Qatar had <24 hours of food and water reserves at peak blockade risk, exposing catastrophic vulnerability.
- *Brine disposal:* desalination generates a concentrated brine (twice as salty as seawater), which

is discharged back into the Persian Gulf. At Qatar's scale, this significantly raises local salinity, damages coral reefs and marine ecosystems, and reduces fish populations. The Gulf is already among the world's saltiest seas; continued brine discharge may reduce the efficiency of future desalination plants as input salinity rises.

— *Climate change*: rising sea temperatures increase the energy required for desalination (warm water is harder to desalinate); intensifying heat waves increase air-conditioning demand and hence water demand simultaneously.

Response strategies: Qatar's National Water Security Master Plan (2016) targets reducing reliance on desalination through: treated wastewater reuse for irrigation (targeting 100% wastewater reuse by 2030); aquifer storage and recovery (injecting treated water into aquifers during low demand for extraction during crises); demand management (metering, pricing reform — though water remains heavily subsidised); and solar-powered desalination pilots. Qatar's wealth means financial barriers are not the constraint — governance, political will, and consumer behaviour change are the challenges.

Germany: The Energiewende

Case Study: Germany — The Energiewende (Energy Transition)

Germany's **Energiewende** ("energy transition") is the world's most ambitious national energy system transformation: a legally mandated transition from nuclear and fossil fuels to renewable energy, combined with radical improvements in energy efficiency. The policy framework includes the Renewable Energy Sources Act (EEG, 2000), the Nuclear Phase-Out (completed April 2023), and the Climate Action Programme.

Key targets and progress:

- Renewables share of electricity generation: 62% in 2023 (up from 6% in 2000). Target: 80% by 2030.
- Wind: 35% of electricity (2023); Germany has 30,000+ onshore wind turbines and is the EU's largest offshore wind producer (second in Europe after the UK).
- Solar: 12% of electricity; Germany is the world's 5th largest solar nation despite its relatively low sunshine hours.
- Nuclear phase-out completed: Last three German nuclear plants (Emsland, Neckarwestheim, Isar 2) closed April 2023, cutting ~6% of electricity supply.

- CO₂ emissions from electricity generation fell 47% between 1990 and 2023.
- Energy efficiency: buildings refurbishment programme, industrial efficiency standards.

The Russia-Ukraine war shock (2022): Russia supplied ~55% of Germany's natural gas, ~35% of oil, and ~20% of coal before the invasion. The 2022 embargo triggered an acute energy crisis: European gas prices rose 400%; Germany was forced to reactivate mothballed coal-fired plants ("Kohlereserve"), temporarily extend some nuclear plants' lives (reversed), and complete emergency LNG terminal construction (6 floating terminals in 9 months — normally a 5-year project). Germany spent €295 billion on energy market interventions and household support in 2022–2023. The crisis brutally exposed Germany's strategic vulnerability from fossil fuel import dependence — the very problem the Energiewende was designed to solve, but had not yet solved.

Structural challenges and criticism:

- *The Dunkelflaute* ("dark doldrums"): periods of days or weeks in winter when both wind and solar produce very little (low sun, no wind). Germany lacks sufficient storage capacity; gas peakers fill this gap, creating continuing dependence on fossil fuels for reliability.
- *Grid infrastructure*: North Germany (wind) needs to transmit power to South Germany (industrial centres) via new high-voltage transmission lines, but planning and permitting processes have delayed construction by 10–15 years. Current transmission constraints mean renewable energy is sometimes curtailed (wasted) because it cannot reach where it is needed.
- *Cost*: German electricity prices for households (€0.34/kWh, 2023) are the highest in Europe, creating fuel poverty pressures and reducing industrial competitiveness. The Energiewende has cost an estimated €500 billion since 2000.
- *Deindustrialisation risk*: energy-intensive industries (steel, chemicals, glass, paper) face higher costs than competitors in countries with cheaper energy. BASF (largest chemical company in Europe) has announced partial relocation to China; ThyssenKrupp is restructuring its steel operations.

Key lessons: Germany demonstrates that the energy transition is technically achievable but geopolitically risky (import dependence), economically costly, logistically complex (grid infrastructure), and politically contentious (nuclear debate, wind turbine siting). It also shows that rapid decarbonisation of electricity is possible through determined policy and sustained investment, but that decarbonising heating, industry, and transport is far more difficult. The

Energiewende remains the world's most important real-world experiment in energy system transformation.

8. Virtual Water and the Food Miles Debate

Virtual Water Trade

Virtual water: The volume of water embedded in — and required to produce — a traded commodity. The concept was developed by John Allan (1998), who used it to explain how Middle Eastern countries had managed to avoid water wars despite extreme scarcity: they imported "virtual water" by buying food produced elsewhere. Importing 1 tonne of wheat (requiring ~1,800 m³ to grow) is equivalent to importing 1,800 m³ of water.

Water intensity of key commodities:

Food item (per kg produced)	Virtual water content (litres)	Implications
Beef	~15,400	1 beef burger requires ~2,400 litres of water. Global shift towards meat-rich diets is a major driver of water demand growth.
Chocolate	~17,196	Cocoa production (West Africa) uses enormous water; drought threatens supply.
Coffee (roasted)	~21,000	Largely rain-fed but sensitive to rainfall patterns; climate change threatens prime growing regions in Ethiopia, Brazil, Vietnam.
Chicken	~4,325	Considerably less water than beef; dietary shift from beef to chicken significantly reduces virtual water demand.
Wheat	~1,827	Major traded virtual water commodity; Ukraine and Russia account for 30% of global wheat exports.

Rice	~2,497	Paddy irrigation accounts for ~40% of global irrigation water; particularly important in water-stressed parts of Asia.
Tomatoes	~214	Very water-efficient; vegetables generally have much lower virtual water than animal products.

Policy implications of virtual water:

- **Water-scarce countries should import virtual-water-intensive goods** (e.g. Qatar, Israel, and Jordan import much of their food rather than growing it domestically, effectively importing water). This is rational resource management.
- **Water-rich countries have a comparative advantage** in producing water-intensive goods; trade allows global water resources to be used more efficiently overall.
- **Limitations:** virtual water calculations assume water has a uniform value regardless of source, but water in a water-stressed region is far more valuable than water in a water-abundant region. Also, some countries "export" virtual water unsustainably (depleting aquifers to grow export crops): e.g. the USA grows almonds in California (the world's most water-stressed agricultural region) for export. Virtual water accounting does not capture water stress, only volume.

Food Miles vs Food Security: The Debate

Food miles: The distance food travels from production to consumer. A high food-miles diet is associated with greater carbon emissions from transport and is criticised as environmentally unsustainable. The UK imports 50% of its food, with average food miles per product increasing significantly since the 1970s as global food systems have become more integrated.

Argument for minimising food miles	Counter-argument: Food miles oversimplify
Transport (aviation, shipping, trucking) generates significant CO ₂ emissions. Air freight (for fresh produce like green beans or strawberries) has a particularly high carbon intensity.	Transport accounts for only 6% of global food system emissions (Poore & Nemecek, 2018, Science). Production method is far more important: a grass-fed lamb from New Zealand transported to the UK has lower total lifecycle emissions than a UK lamb raised in an energy-intensive shed.

Local food supports local farmers, preserves local agricultural landscapes, and shortens supply chains (reducing waste and improving freshness).	Seasonal local food may have higher emissions than out-of-season imports: UK tomatoes grown in heated greenhouses in winter use more energy than Spanish tomatoes trucked 2,000 km.
Long global supply chains are vulnerable to disruption (COVID, conflict, extreme weather). Food sovereignty (ability to feed a population from domestic production) is a security concern.	Global trade enables comparative advantage: countries produce what they can grow efficiently (climate, soil, water permitting), improving overall resource efficiency. Trade also stabilises supply by diversifying sources.
Fair trade issues: some "local" food advocates argue that buying local displaces imports from LIC farmers who depend on export earnings for development.	Fair trade certification shows that buying from LIC farmers (e.g. Kenyan green beans, Ethiopian coffee) provides crucial income for small farmers and supports development goals. Refusing imports harms producers.
Urban food growing and urban agriculture can produce fresh food with minimal food miles and strengthen community resilience.	Urban agriculture cannot feed cities at scale: even if every available space in London was cultivated, it could provide <5% of London's food needs. It has social and educational value but cannot replace conventional agriculture.

Food sovereignty: A concept promoted by La Via Campesina (international farmers' movement) asserting that peoples and communities have the right to define their own food and agriculture policies, produce their own food using their own methods, and prioritise local food production over export-oriented agriculture. Food sovereignty challenges the neoliberal food miles debate by centring the rights of producers (especially smallholders) over the preferences of consumers in wealthy countries.

Exam Tip: The food miles vs food security debate is a strong AQA-style evaluative question: "Assess the view that reducing food miles is the most effective strategy for improving food security." Structure: (1) argument that reducing food miles builds local resilience and reduces emissions; (2) counter-argument that production method matters more than distance; (3) fair trade and LIC development considerations; (4) virtual water trade as an alternative framework; (5) conclusion: most effective strategies address production methods, food waste, dietary change

(reducing meat), and equitable distribution alongside (not instead of) more local production where appropriate.

9. The Global Energy Mix: Detailed Analysis

Energy Source	% Primary Energy 2023	% Electricity 2023	Key Producers	Key Considerations
Oil	31%	3% (declining)	USA (18%), Saudi Arabia (12%), Russia (12%)	Dominant for transport (56% of oil use is road transport). Petrochemicals (plastics, fertilisers) ensure continued demand even as EVs displace road use. OPEC+ manipulates price via production quotas. Carbon content: ~0.73 kg CO ₂ /kWh.
Coal	26%	36%	China (55% of global), India (10%), USA (9%)	Cheapest source of baseload electricity in many LICs/MICs. Highest carbon intensity: ~1.0 kg CO ₂ /kWh. China added 200 GW of new coal capacity in 2022 despite climate pledges. G7 target to phase out unabated coal by 2030 (EU) or mid-2030s (UK).
Natural Gas	23%	23%	USA (22%), Russia (15%), Iran (6%)	Rapidly growing as coal replacement. LNG (liquefied natural gas) trade expanding, reducing pipeline dependency. Carbon content: ~0.49 kg CO ₂ /kWh (half of coal). Methane leakage from infrastructure may offset climate benefit. Shale gas revolution

				(fracking) made USA world's top gas producer by 2017.
Hydroelectric	7%	15%	China (31% of global capacity), Brazil (9%), USA (7%)	Zero operational emissions; reliable baseload power. Limited new sites available in HICs. Grand Ethiopian Renaissance Dam (GERD, 6,000 MW) shows LICs expanding capacity. Vulnerable to drought (Brazil's hydro output fell 30% during 2021 drought).
Nuclear	4%	10%	USA (31%), China (15%), France (15%)	Zero operational carbon; reliable baseload. France: 70% nuclear electricity (most nuclear-dependent nation). Japan restarted reactors post-Fukushima slowly (from 0 in 2012 to 10 by 2024). High construction cost/time; waste storage unresolved; public acceptance declining. Small Modular Reactors (SMRs) may change economics.
Wind	4%	8%	China (40% of global capacity), USA (16%), Germany (8%), UK (6%)	Onshore wind now cheapest new electricity source in most countries (~£30–50/MWh). Offshore wind more expensive but higher capacity factor. UK: 27% of electricity from wind (2023); 14 GW offshore. Intermittency requires storage or backup. NIMBY opposition to onshore wind in UK cost decade of progress (ban 2015–2023).
Solar PV	2%	5%	China (36% of generation), EU (9%), USA (5%)	Fastest-growing source: costs fell 90% 2010–2023. Now cheapest new electricity in most of the world (<\$20/MWh in some sun-belt locations). China produces 80% of

				global solar panels. Intermittency (night, cloudy weather) is the fundamental challenge; battery storage costs also falling rapidly.
Bioenergy	~3%	2%	USA (12%), Brazil (9%), EU (25% of EU renewable energy)	Includes traditional biomass (cooking fires, 700M people) and modern bioenergy. Drax (UK) converted from coal to wood pellets (imported from US/Canada); claimed as "renewable" but controversial (burning trees releases stored carbon immediately; forest regrowth takes decades). Biofuels compete with food for land.

Regional energy security profiles:

- **Middle East and North Africa (MENA):** World's largest oil and gas reserves (60% of proven oil; 40% of gas). Paradoxically, several MENA countries face high domestic electricity costs and are investing in solar. Saudi Arabia's NEOM project aims to run entirely on renewables; UAE already has world's largest solar farm (Noor Abu Dhabi, 1.2 GW).
- **Sub-Saharan Africa:** 600 million people lack electricity (World Bank, 2023). Vast renewable potential (Congo River hydropower: theoretical 100,000 MW; Saharan solar). But grid infrastructure, investment, and governance are the barriers, not resource availability. The debate: large hydropower (GERD, Grand Inga) vs. distributed solar (rooftop solar, mini-grids). Africa receives only 2% of global clean energy investment despite having some of the world's best renewable resources.
- **East Asia:** China and Japan represent opposite ends of the energy transition spectrum. China: massive simultaneous investment in renewables AND coal (energy security/development priority). Japan: nuclear-scarred (Fukushima 2011) but energy import-dependent; gradual nuclear restart and offshore wind expansion.

Exam Tip: Resource security questions in AQA often present conflicting imperatives. For example: "Evaluate the extent to which renewable energy can ensure energy security." Arguments FOR: inexhaustible supply, domestic production reduces import dependency, falling costs, zero carbon. Arguments AGAINST: intermittency requires backup/storage, infrastructure gaps, not all countries have good resources, transition requires decades, mining of critical minerals creates new

dependencies, and renewable energy doesn't solve water or food insecurity. A strong conclusion: renewables are essential for long-term energy security but cannot alone provide the full security package in the short-to-medium term without complementary storage, grid investment, and demand management.

Exam Tip: Always use specific percentage figures and data in resource security answers: "coal provides 26% of global primary energy"; "Saudi Arabia desalinates 50% of its water supply"; "the UK's offshore wind capacity of 14 GW is the world's second largest." Quantitative precision distinguishes high-quality answers. Memorise key figures for oil, coal, gas, renewables shares of the energy mix, water stress thresholds, and food insecurity statistics from the FAO.